

Example Lecture – Hamilton's principle

For your **20min** teaching presentation we would ask you to give part of a lecture for the 2nd year course on **Classical Dynamics** that forms part of IB Physics B, i.e. the 2nd year of physics within the natural sciences tripos. You can find more information about the course here:

<https://www.phy.cam.ac.uk/students/teaching/current-courses>

This lecture would be in the middle of the Classical Dynamics course, i.e. the previous lectures will have covered the syllabus copied below up to the point marked in yellow.

You are to provide a 20min introduction on **Hamilton's principle (principle of stationary** , which should include:

- Definition of Action and a discussion of its physics significance
- Definition of Lagrangian and a discussion of its physics significance
- Outlook: Neat applications or connection to other parts of physics

The teaching presentation will happen in person, in a room equipped with a whiteboard/blackboard, a projector and Wifi. Since the presentation will take place out of term time, attendance will be rather low and not contain many actual students. In real life, however, the audience will consist of >200 students, of which about 75% will continue into part II physics, and you can indicate during your lecture how you would engage / interact with a cohort of this size.

Please direct any questions you might have to the Director of Undergraduate Education in Physics, Prof Ulrich Schneider (uws20@cam.ac.uk).

Part IB Classical Dynamics

Prerequisites

This course builds on the ideas introduced in NST Part IA “Physics”, using the tools of vector calculus taught in NST Part IA “Mathematics”.

Learning Outcomes and Assessment

The main areas covered are orbits, rigid body dynamics, normal modes and continuum mechanics (elasticity and fluids). However, the real purpose of this course, as with so many other courses in 1B Physics, is to demonstrate the enormous range of phenomena that are all explained by single set of fundamental laws or equations of Physics - in this case, Newton's Laws of Motion.

Synopsis

Newtonian mechanics, frames of reference: Review of Part IA mechanics: many-particle system, internal and external forces and energy. Central forces, motion in a plane. Non-inertial frames, rotating frames, centrifugal and Coriolis forces.

Orbits: Effective potential and radial motion, bound and unbound orbits. Inverse-square law orbits, circular and elliptic, Kepler's laws. Escape velocity, transfer orbits, gravitational slingshot. Hyperbolic orbits, angle of scattering, repulsive force. Two-body problem, reduced mass. General features of three-body problem. Brief treatment of tidal effects in gravitational systems.

Rigid body dynamics: Instantaneous motion of a rigid body, angular velocity and angular momentum, moment of inertia tensor, principal axes and moments. Rotational energy, inertia ellipsoid. Euler's equations, free precession of a symmetrical top, space and body frequencies. Forced precession, gyroscopes.

Introduction to Lagrangian mechanics: Generalised coordinates. Hamilton's principle and Lagrange's equations. Symmetries and conservation laws. Conservation of the Hamiltonian for time-independent systems.

Normal modes: Analysis of many-particle system in terms of normal modes. Degrees of freedom, matrix notation, zero-frequency and degenerate modes. Motion in three dimensions, modes of molecules.

Elasticity: Hooke's law, Young's modulus, Poisson's ratio. Bulk modulus, shear modulus, stress tensor, principal stresses, strain tensor. Elastic energy. Torsion of cylinder. Bending of beams, bending moment, boundary conditions. Euler strut. Brief treatment of elastic waves.

Fluid dynamics: Continuum fields, material derivatives, relation to particle paths and streamlines. Mass conservation, incompressibility. Convective derivative and equation of motion. Bernoulli's theorem, applications. Velocity potential, applications: sources and sinks; flow past a sphere and cylinder; vortices; Magnus effect. Viscosity, Couette and Poiseuille flow. Reynolds number, laminar and turbulent flow.

Earlier courses:

Part IA Dynamics

Prerequisites

This course assumes knowledge of the material in A-level Physics and Mathematics; however it is also accessible to students who have not taken A-level Physics but who have taken three Mechanics modules in A-level Mathematics and Further Mathematics.

Learning Outcomes and Assessment

In this course you will be introduced firstly to important concepts in the collection and analysis of experimental data and secondly to the application of rigorous mathematical models to describe familiar concepts in statics and dynamics. In addition you will be learning more sophisticated ways of tackling physics problems. We will work through numerous examples in the lectures which illustrate the physics and mathematics you are learning and show you how to think about and work through such questions.

At the start of the course you are given an introduction to university physics and will gain an understanding of the scientific method and the role and importance of both experiment and mathematical modelling in forwarding our understanding of the physical world. You are introduced to the use of dimensional analysis as an important tool in understanding the relationships between the different physical properties of a system. The differences between the prescriptive step-by-step approach taken to tackling Physics problems at A-level and the much less structured way of tackling physics problems at university are highlighted.

The two lectures on experimental physics are designed to fit in with what you are learning in the practical classes. You will understand the concept of a random error and be introduced to the Gaussian probability distribution - on which the analysis of the significance of errors is predicated. You will learn about the importance of making several measurements to improve the accuracy of your result and how to calculate the mean, standard deviation and error in the mean for your data. You will also learn about how to find the errors in functions of a single variable and also how to combine errors in two or more variables. You will understand the concept of a systematic error and learn about some techniques for dealing with systematic errors - with particular reference to the experiments carried out in the practical class.

In the sections on statics and dynamics you will be reacquainted with various concepts you have already met but cast in a more rigorous mathematical way. You learn about how to use calculus in physics in a simple way and will understand the basic application of differentiation and integration in analysing physical models; you will learn to set up differential equations (DEs) to describe the motion of physical systems - solving the equations directly in the case of first-order DEs but simply using substitution of a suggested solution in the case of second order DEs. Another useful

mathematical tool you will learn is that of approximation i.e. looking at the behaviour of an equation or solution in the limiting case when the variable is either very small or very large.

You will understand the concept of a force: as something that has a tendency to produce motion and that is a vector which has components which can be resolved in different directions; you will understand the concept of equilibrium and the value of using free-body diagrams to analyse the forces acting on a system. The forces exerted by idealised springs, strings and pulleys, and the effects of friction on systems, will be considered. The concept of the work done by a force is introduced in 1-D and in 3-D using scalar (dot) products, and using integration in the case of a force which is a function of position; this leads on to the definition of potential energy, and its relationship to the internal forces in a system and to stable and unstable equilibrium.

The relationships between displacement, velocity and acceleration are described using calculus. You will look at Newton's 1st and 2nd laws of motion and use the 2nd law to derive the differential equations (the equations of motion) describing various physical systems. The rate of doing work on a system (the power) is described as the derivative of the work done. The kinetic energy of a system is defined through considering the work done accelerating it. You will learn about the principle of the conservation of energy - specifically that in an isolated system energy in all its forms is conserved. The concept of momentum is introduced and through considering Newton's 2nd law you will learn that the force applied equals the rate of change of momentum. From this and Newton's 3rd law it is shown that in an isolated system linear momentum is conserved. This law is used to analyse the motion of rockets and to look at elastic and inelastic collisions between two particles. The impulse of a force is defined as the integral of a force over the period for which it acts and is equal to the change in the momentum of the associated system.

You will understand the concept of a frame of reference, with particular reference to inertial frames of reference i.e. those which are travelling at a constant speed with respect to one another, and learn how to transform coordinates from one frame to another. In particular you will learn how to use a zero-momentum frame which can greatly simplify the analysis of collisions between bodies.

Synopsis

Introduction to university physics: role of experiment; mathematical models; dimensional analysis; tackling physics problems.

Experimental physics: random and systematic errors; Gaussian probability distribution; mean, standard deviation, error in the mean; errors in functions of a single variable, combining errors in two variables; examples of techniques for dealing with systematic errors; graphs.

Dynamics: *Concept of a force:* tendency to produce motion; forces as vectors; action and reaction; friction. *Calculus in physics:* use of integration. *Work:* potential energy; stable and unstable equilibrium. *Kinematics:* displacement, speed, velocity, acceleration. *Newton's laws of motion:* equations of motion. *Kinetic energy:* concept and definition; principle of the conservation of energy. *Linear momentum:* concept and definition; conservation of linear momentum; rockets; elastic and inelastic collisions; impulse of a force. *Frames of reference:* relative velocities, inertial frames of reference, zero-momentum frame, collisions.

Part IA Rotational Mechanics and Special Relativity

Prerequisites

Pre-requisites for IA Physics are that students have studied maths for public examination at A level or equivalent and physics or the mechanics modules of further maths.

For this course it is assumed that students have completed the IA Physics dynamics course in the Michaelmas term.

Learning Outcomes and Assessment

At the end of this course students should be able able to:

(Rotational Mechanics):

1. Combine concepts from the first term course in mechanics be able **to solve problems for extended bodies** that are in equilibrium or accelerating using
 - a) Newton's Laws of motion linearly and rotationally
 - b) Conservation of momentum and angular momentum
 - c) Conservation of energy linearly and rotationally
 - d) The equations describing angular and circular motion
2. To understand and calculate the constant precession of a gyroscope.

(Special Relativity):

Primary objectives:

1. Describe and explain the **experimental evidence** for special relativity.
2. Derive and use the **Lorentz transformations** to relate, for example, displacement, time, velocity, momentum and energy in frames moving relative to one another at relativistic speeds.
3. Apply the **energy-momentum invariant** and **relativistic energy and momentum equations** to solve problems of particle collisions.

Secondary objective:

4. Draw **time-distance diagrams** to describe and understand relativistic problems in different frames of reference (e.g. the Earth and rocket frames)

Synopsis

Rotational Mechanics:

Turning moments: lever balance; turning moment as a vector; moment of a couple; conditions for static equilibrium.

Centre of mass: calculation for a solid body by integration.

Moment of inertia: calculation of moment of inertia; theorems of parallel and perpendicular axes.

Circular motion: angle, angular speed, angular acceleration; as vectors; rotating frames; centripetal force.

Angular momentum: concept and definition; angular impulse; conservation.

Rotational kinetic energy: simple collisions involving angular rotation.

Gyroscope: how it works; precession.

Special Relativity:

Historical development: problems with classical ideas; the Aether; Michelson-Morley experiment.

Inertial frames: Galilean transformation.

Einstein's postulates: statement; events, and intervals between them; consequences for time intervals and lengths;

Lorentz transformation of intervals: simultaneity; proper time; twin paradox; causality; world lines and space–time diagrams.

Velocities: addition; aberration of light; Doppler effect.

Relativistic mechanics: momentum and energy; definitions; what is conserved; energy–momentum invariant. Nuclear binding energies, fission and fusion.